

Oil Price Shocks, Competition, and Oil & Gas Stock Returns- Global Evidence

Kartick Gupta

University of South Australia

37 North Terrace

Adelaide 5000, Australia

Email- Kartick.gupta@unisa.edu.au

Telephone- +61 8 83027179

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Abstract

Extant literature suggests that oil price shocks have a strong impact on the macro economy and the stock market. However, relatively less is known about the effect of country-level determinants, competition, and asymmetrical relationship in affecting the oil & gas returns at the firm-level. Using comprehensive firm-level monthly data from 72 countries spanning 1983 to 2014, we find: (i) macroeconomic stress negatively impact firm-level returns; (ii) oil price shocks positively impact firm-level returns; (iii) firms located in high oil producing countries are more sensitive to global uncertainty and oil price shocks; (iv) firms located in non-competitive industries are less sensitive to oil price shocks; (v) firms located in non-competitive industries are less affected by the drop in oil price, as compared to firms that are located in highly competitive industries.

JEL Classifications: C23; G12; Q43

Keywords: Equity returns; Oil prices; Market dislocation; Production and consumption; Competition

1. Introduction

To what extent are the systematic asset price risk of oil & gas firms explained by the oil price shocks that are themselves affected by macroeconomic forces? Are monopoly firms less affected by the volatility in the oil prices? Does the stock market react symmetrically to the oil price increase or decrease, especially when the competition level is taken into consideration? We attempt to answer these important issues in this paper. We find that including a measure of macroeconomic stress can better explain the systematic asset price risk of the oil & gas firms. The firms that are located in high oil producing countries or face strong competition from other firms are more sensitive to oil price shocks and market stress. Finally, monopoly firms are less affected by the drop in oil price than firms that face strong competition from other firms.

The relationship between oil price and economic activity has been extensively studied in the extant literature. The general consensus is that oil price shocks can significantly affect the economic activity of the country. For example, Hamilton (1983) find that oil price shocks are responsible for majority of the US recession, post-World War II. The findings are further tested and supported by Burbidge and Harrison (1984) and Gisser and Goodwin (1986). Recently, a number of studies have extended the relationship between oil prices and economic activity to the stock market. The evidence, however, is mixed. For example, Jones and Kaul (1996) and Sadorsky (1999) find a negative relationship between oil price shocks and stock returns, while Chen et al. (1986) and Huang et al. (1996) do not find a significant relationship between oil prices and the US stock return.

Sadorsky (2001) and Hamilton (2003) study the sources of oil price shocks and indicate that the oil prices may themselves be affected by macroeconomic forces. This suggests that oil price shocks may not be exogenous as macroeconomic forces, such as political, economic and financial risks affect systematic asset price risk. For instance, the underlying reason for the

volatility in oil price was due to political (1990 Gulf War), economic (1994 Mexico; 1997 East Asia crisis; 2001 Argentina), or financial reasons (1998 LTCM Russia; 2008 Mortgage-backed-securities US) [See Figure 1 & 2]. Further, financial innovation in exotic securities have led to more trading in oil futures, derivatives, and other similar instruments. Therefore, market stress and disruption can adversely impact trading of these instruments, leading to distortion of oil prices (Wen et al., 2012).

A few papers addressed this issue by considering macroeconomic factors in addition to oil price shocks. For example, Sadorsky (2001) and Boyer and Filion (2007) include USD/CAD exchange rates and interest rates to control macroeconomic determinants. However, the USD/CAD exchange rate and the interest rate prevalent in the local market may not fully capture the extent of global market stress or uncertainty. To this end, we use a better measure of macroeconomic stress (market dislocation index) proposed by Pasquariello (2014). The market dislocation is defined as the stressful conditions in which asset mispricing increases significantly. Pasquariello (2014) notes that market dislocation measures are reasonable proxy for a range of international, economic and financial market variables, such as NBER recession dummy, changes in Volatility Index (VIX), world market return volatility, “TED”, and slope of yield curve spread, among others. As seen in Figure 2, this measure tracks important events of the macro economy, such as the European currency crisis in 1992–1993, the Asian crisis in 1997, the Russian default in 1998, the Global Financial Crisis (GFC) of 2008–2009, the European debt crisis of 2009-2010, and the Arab Spring in 2011-2012. Therefore, we argue that combining oil price shocks and market dislocation can better explain the systematic asset price risk of oil & gas firms.

The impact of oil price shock on firm-level returns in the presence of product market competition requires further investigation. Increasing rates of mergers and acquisitions in the US oil companies has resulted in consolidation and higher wholesale gasoline prices

(Oladunjoye, 2008). Consequently, an increase in competition may result in sticky gasoline prices and therefore the firms may cushion themselves from oil price shocks. Although the extant literature (see Deltas, 2008; Oladunjoye, 2008 among others) study the effect of gasoline prices in the local market in the presence of competition, a comprehensive analysis has been limited to date. Most of the studies are restricted towards the gasoline prices at the retail and wholesale level and not much is known on how the effect will be transmitted to the stock market.

The role of country-level determinants in affecting firm-level oil & gas stock returns requires further analysis. It can be argued that the performance of oil & gas firms depend on country-level determinants, such as production and consumption of oil. For example, Russia, Venezuela, Saudi Arabia and other oil-exporting countries generate a large share of revenue from oil exports and therefore a drop in oil prices can have a major impact on the overall economy. Lopez-Murphy and Villafuerte (2010) find that a small decrease in oil prices increases large demand for finances in the near term for oil producing countries. Further, falling oil prices were some of the primary reasons for the Russian Government to devalue its currency and default on debt (Sadorsky, 2001). An investigation of how the stock market responds to oil price shocks, taking into account country-level determinants may provide additional evidence on the importance of country-level determinants.

Previous studies frequently use sub-index as a benchmark to assess the impact of oil prices on the stock market. However, it is well known that the use of indices introduce a number of shortcomings that can bias the results (Banz and Breen, 1986; Brown et al., 1992). Specifically, Brown et al. (1992) assert that including only surviving stocks in a sample may lead to artificial performance persistence and therefore statistical tests may be unreliable. Further, by the measure of construction, sub-index composites do not capture well the performance of distressed, dead, or delisted stocks. This may artificially inflate the sub-index returns,

especially during the period of market stress when the bankruptcy filing rate rises. In addition, firm-level determinants cannot be controlled if sub-index is used as a performance benchmark. This leads to exclusion of important firm-level determinants, such as leverage, expected cash flows, size, liquidity, discount rate etc., from the estimation procedure (Miller and Ratti, 2009). We address these shortcomings by using firm-level data.

The generalizability of findings from prior studies that mainly investigates industrialized economies also needs further validation. Park and Ratti (2008) study the relationship between oil price shocks and stock returns in the US and 13 European countries. Papapetrou (2001) studies the oil price shocks in Greece; Sadorsky (2001) and Boyer and Filion (2007) investigate the Canadian market, and El-Sharif et al (2005) concentrate on the UK market. However, not much is known on the impact of oil price shocks in firms located in developing and transitional economies. By extending the analysis to 72 countries, we provide evidence on whether oil price shocks are systematic across countries.

Our study contributes to the existing literature in a number of ways. As pointed out by Huang et al. (1996), oil or oil by-products are some of the main input factors that influence profitability and expected cash flows of a firm. Therefore, we use firm-level oil & gas data to improve the robustness of the result. By studying the systematic risk of macroeconomic forces at the firm-level stock returns, covering developed, developing, and transitional countries over a long period, our study contributes to the extant literature that mainly uses sub-index as a performance benchmark.

Second, to the best of our knowledge we are not aware of any other paper that investigates the role of competition within the oil & gas industry in driving firm-level returns. The results from this paper may be of particular interest to policy makers and regulators as the results suggest that monopoly firms are less sensitive to oil price shocks.

Third, our study indicates that country-level determinants are important factors in explaining firm-level returns. Particularly, we show that the firms located in oil-rich countries are more sensitive to oil price shocks and market stress. Further empirical work can explore other country-level determinants that are important in explaining firm-level returns.

The rest of this paper is organized as follows: The background literature is reviewed in Section 2. Data and Methodology is presented in Section 3. In Section 4, we present empirical results and their implications. Finally, Section 5 summarizes research findings and concludes.

2. Background Literature

Prior evidence on the linkage between oil price shocks and financial markets come from (Huang et al., 1996; Jones and Kaul, 1996; Sadorsky, 1999). Particularly, Sadorsky (1999) shows the importance of oil prices and their volatility in explaining real stock returns. Huang et al. (1996) report contradictory findings as the authors fail to find any strong connection between oil futures and aggregate-level market return. Elyasiani et al. (2011) extend the analysis to 13 US indices and find that oil price shocks and oil returns volatility constitute systematic asset price risk for nine out of 13 industries. Kilian and Park (2009) study the sources of oil price shocks and document that demand and supply imbalance impact stock returns differently. However, the authors only consider the demand and supply imbalance but not other factors that can lead to oil price shocks.

Studies have been also undertaken in markets outside the US. Using aggregate level oil & gas sub-index based in Canada, Sadorsky (2001) finds a positive relationship between oil price and oil & gas stock prices; while a negative relationship is observed between exchange rate, term premium, and oil & gas stock prices. El-Sharif et al. (2005) extend the analysis to the UK market and consistent with Sadorsky (2001), the authors find a positive relationship between oil price and oil & gas stock prices. Boyer and Filion (2007) re-examine the determinants of

oil & gas returns in the Canadian market by considering five common and five fundamental factors. Using firm-level data, the authors show how each of the common and fundamental factors affects oil & gas stock returns. The authors assert that increase in oil prices have a positive effect on the oil & gas stock prices.

Nandha and Faff (2008) analyse the relationship between oil and global industries. Using 35 Datastream global industry indices that represent the performance of each industry across countries (instead of each country), the authors note that an increase in oil prices adversely affects the returns of all industries except mining, and oil & gas industries. Park and Ratti (2008) analyse the impact of oil price shocks on real stock returns in the US and 13 European countries between 1986 and 2005. The authors confirm that oil price shocks explain the variability of real stock returns in the same month or within a month.

3. Data and Methodology

3.1. Market Dislocation Index (MDI)

We follow Pasquariello (2014) in constructing the market dislocation index. The market dislocation measure is based on the law of one price and derived from the deviation in the Covered Interest Rate Parity (CIRP), Triangular Arbitrage Parity (TAP) and American Depositary Receipt Parity (ADRP). Since the approach does not depend on any model, this significantly reduces model misspecification and as a result increases the robustness of the measure. Pasquariello (2014) states that the MDI captures market conditions that are under stress and experiencing large, widespread asset mispricing. We briefly outline the construction approach as following:¹

¹ See Pasquariello (2014) for detailed MDI construction procedure.

3.1.1. Covered Interest Rate Parity

As per CIRP law of one price, the theoretical forward exchange rate of borrowing in any currency A at the interest cost r_A for $T-t$ days, exchanging the borrowed amount to currency B at the spot exchange rate $S_{t,A/B}$ and depositing in currency B at interest rate r_B is as following:

$$F_{t,T,A/B}^* = S_{t,A/B} \left(\frac{1+r_{A,t,T}}{1+r_{B,t,T}} \right)$$

Therefore, the law of one price holds the difference between market forward exchange rate between currency A and B ($F_{t,T,A/B}$) and the theoretical forward exchange rate ($F_{t,T,A/B}^*$) should be close to zero.

In order to calculate daily CIRP violations, we use the same currency suggested by Pasquariello (2014). We retrieve daily spot and forward exchange rates of the five most heavily traded foreign exchange currencies from 1st May, 1990 to 31st December, 2014.² These are CHF/USD, GBP/USD, EUR/USD, JPY/USD, CHF/EUR, GBP/EUR, JPY/EUR, CHF/GBP and JPY/GBP. We also retrieve corresponding LIBOR rates of each country for seven different maturities (7, 30, 60, 90, 180, 270, and 360 days). Nine different exchange rates with seven different maturities results in 63 CIRP permutations. Both the exchange rates and LIBOR are available from Datastream. Finally, for each permutation we compute absolute log differences as, $CIRP_{i,t} = |\ln(F_{t,T,A/B}) - \ln(F_{t,T,A/B}^*)| * 10,000$.

3.1.2. Triangular Arbitrage Parity

The second arbitrage violation deals with price discrepancy among three different currencies. This approach entails buying the vehicle currency indirectly through other cross-rates currency.

² Forward rates prior to 1st May 1990 are unavailable.

For example, instead of buying USD/JPY directly, an investor can buy USD/GBP and then exchange with GBP/JPY, resulting in USD/JPY. In the absence of arbitrage opportunity, an investor will find the difference between direct and cross-currency exchange rate close to zero, i.e., $S_{t,A/B} - S_{t,A/B}^* = 0$, where $S_{t,A/B}^* = S_{t,A/V} * S_{t,V/B}$, V is the vehicle currency and, A and B are the cross-rate currencies.

We retrieve daily spot exchange rates of the most liquid, relatively free-floating currencies of AUD, CAD, CHF, FRF, GBP, ITL and JPY. We use either USD or EUR as a vehicle currency.³ This dataset comes from the Pacific Exchange Rate Service. For each of the 112 TAP permutations, we compute daily absolute log differences between direct currency and cross-rates as, $TAP_{i,t} = |\ln(S_{t,A/B}) - \ln(S_{t,A/B}^*)| * 10,000$.

3.1.3. ADR Parity

The last set of arbitrage deviation relates to price discrepancy between the stock listed in the domestic market and US-listed American Depositary Receipt (ADR), taking into account bundling ratio and corresponding date exchange rate. The theoretical ADR price derived from the domestic listed stock is as following:

$$P_{i,t}^* = S_{t,USD/H} * q_i * P_{i,t}^H$$

where $S_{t,USD/H}$ is the spot exchange rate between local currency and USD, q_i is the bundling ratio and $P_{i,t}^H$ is the unit stock price of the underlying foreign shares in their local currency H . In the absence of trading frictions, an arbitrage opportunity may exist if $P \neq P_{i,t}^*$. In order to find ADRP violations, we include 247 domestic-U.S. pairs of daily closing stock prices from 40

³ We use DEM instead of EUR as one of the vehicle currencies prior to the introduction of the Euro. Similarly, we use FRF and ITL as cross-rate currencies prior to the introduction of the Euro.

countries that are listed as ADR in either NYSE, NASDAQ or AMEX stock exchange. For each pair, we calculate ADRP violation as $ADRP_{i,t} = |\ln(P_{i,t}) - \ln(P_{i,t}^*)| * 10,000$.

The final index construction entails a number of steps as underlined by Pasquariello (2014). First, each parity is standardized relative to its own historical distribution. Next, an equal-weighted average of all three parities is computed on that day. In the last step, the monthly market dislocation index is computed from daily data by taking the average over the month.

3.2. Stock Price

The stock price data is from Datastream. We use International Classification Benchmark (ICB) to find stocks that operate in the oil & gas industry.⁴ This results in retrieving information of 2,379 active and dead/delisted stocks from 72 countries. Consistent with Nandha and Faff (2008) and Park and Ratti (2008), we use monthly frequency data. As discussed earlier, we use firm-level returns instead of sub-index returns to measure the sensitivity of stock returns towards oil price shocks and market uncertainty. The monthly log firm-level return is calculated as $R_{i,t} = \ln(P_t/P_{t-1})$, where P is the price of stock i in month t . The stock price is adjusted for theoretical growth in value of the dividend, assuming re-investment. Thus, the stock prices reflect both capital gain and dividend yield.

3.3. Control Variables

In order to control the effect of other variables that are known to affect the stock return, we use overall market return, illiquidity, standard deviation of stock return, leverage, log of market value and price to book value ratio. We calculate illiquidity as per the Lesmond et al. (1999)

⁴ In unreported results, we consider using industry classification provided by FF48 classification (Fama and French, 1997), which in turn is drawn from the Standard Industrial Classification (SIC). This does not alter our main findings.

model, where a stock with no change in price over a period of time is considered illiquid. This measure is calculated as the ratio of zero trading days to the total number of trading days over the last one year. As a priori, it is expected that illiquidity and stock returns have a positive relationship. Riskiness of firm is computed as daily standard deviation of stock returns over the last one year. It is expected that a risky stock command a higher return. We calculate leverage as long-term debt divided by the total assets. Fama and French (1992) document a negative relationship between firm size and price to book value with stock returns. Consequently, we use the log of market value and price to book value ratio to control these factors. To improve the quality of the data, we follow Ince and Porter (2006) and Chui et al. (2010) to address the data issues noted in Datastream. The control variables and the stock returns are winsorized at 1 and 99 percentile to minimize the effect of outliers.

3.4. Oil Price

We use New York Mercantile Exchange (NYMEX) roll-on-future one month oil price as a proxy for oil & gas prices. Our choice for NYMEX is motivated by Elyasiani et al. (2011) where the authors suggest a number of advantages of using future oil prices over spot prices, including less random temporary noise. Substituting NYMEX with West Texas Intermediate Cushing (WTI) or the UK Brent crude oil does not alter our findings. Following Sadorsky (2001) and Elyasiani et al. (2011) we use risk-free rate adjusted stock and market return. The country-specific market return and risk-free rate are retrieved from Datastream. NYMEX prices are available from the US Energy Information Administration (USEIA) website. The USEIA provides NYMEX prices, starting 29th April, 1983, and therefore our sample period is from May 1983 to December 2014.

3.5. Product Market Competition

We construct sales Herfindahl-Hirschman Index (HHI) as a proxy for product market competition. This measure is extensively used in the extant literature, particularly in the industrial organization literature (see Tirole, 1988 among others). It is calculated by squaring the market share of each firm in the industry within the country and then adding the resulting numbers. The range of HHI measure lies between zero and one, with a high HHI index suggesting monopoly in the given industry. We calculate the sales HHI of at least four firms in the industry out of a maximum number of firms of 50. This is consistent with the US Census that uses the 50 largest firms within each industry or the largest four-firm concentration ratio. We also consider alternatives where the maximum number of firms used in calculating sales HHI is not imposed or only the top four sales firms within the oil & gas industry are considered. The results remain qualitatively the same in either case. We use the Worldscope database to obtain sales figure of publicly listed oil & gas firms. The competition measure is calculated for the oil & gas industry in each country and each year.

Table 1 reports for each sample country, the number of firm-month and country-month observations used in the analysis. It also reports the median oil production and consumption of each country, collected from the USEIA website. A comprehensive 275,865 firm-month observations from 72 countries are used for empirical analysis. The sample has a wide coverage of firms located in developed, developing and transitional economies, including major countries across all geographical regions. We also note that, as expected, the sample is not evenly distributed across countries- Canada and the US dominate the sample. However, robustness checks do not suggest the dominance of firms listed in Canada and the US driving the results. In the fourth and fifth columns, we report the median oil supply and consumption, measured in thousand barrels per day. Saudi Arabia, Russia and the US rank high in oil supply, whereas the US, China and Japan rank high in oil consumption.

[Insert Table 1 here]

We report summary statistics of our control variables in Table 2. We find the average monthly stock return is -0.56% with a standard deviation of 20.89%. Compared to stock return, average market return is 0.60% with a reasonable standard deviation of 5.20%. We also document bias that may arise if sub-index returns are used instead of firm-level returns. In untabulated results, we calculate oil & gas sub-index return and find average and median monthly return of multi-country oil & gas index as 0.64% and 0.30% respectively. The standard deviation of sub-index return is 8.09%, suggesting that sub-index returns are positively skewed and less volatile than firm-level returns. We also find that out of a total of 2,370 firms, only 607 were operating at the end of 2014. This indicates that close to 75% of the firms are delisted due to various reasons. As seen in Table 2, the illiquidity of oil & gas firms is 39.06%, suggesting that on average the prices of firms did not change in four out of 10 days. Similarly, the average and median market valuation is 38 and 33 million USD respectively. These results jointly suggest that the performance of the majority of the firms may not be captured by an aggregate level sub-index.

[Insert Table 2 here]

In Table 3, we report correlation matrices between dependent and independent variables. The correlation between stock and market return is 0.213 and statistically significant at the one percent level. As oil prices positively impact oil & gas stock returns, we find a correlation of 0.141 between stock and NYMEX return. MDI is negatively correlated with stock return, suggesting that an increase in global market stress decreases the return of oil & gas firms. Firm

size is positively related to the stock return. We also do not find the problem of multicollinearity as none of the correlations exceeds absolute 0.50.

[Insert Table 3 here]

4. Empirical Results

Before undertaking further analysis, we verify if the data suffers from a problem of unit root. In Table 4, we present a modified Dickey-Fuller t -test, proposed by Elliott et al. (1996) to check if the unit root problem exists. The results indicate that the MDI index is stationary as the DF-GLS τ -statistic is outside of the one percent critical value in all four lags and therefore the null hypothesis of unit root is rejected. However, oil price (NYMEX) τ -statistic is within the one percent critical value and therefore the series suffers from unit root problem. Using the log difference (returns) of NYMEX, though, solves the problem as the null hypothesis of unit root is rejected at the one percent level. In unreported results, we find that unit root tests suggested by Phillips and Perron (1988) and Kwiatkowski et al. (1992) arrive at the same outcome. We also find (unreported) that other variables used in the analysis follow $I(0)$ processes.

[Insert Table 4 here]

In Table 6, we investigate the impact of oil returns and market dislocation on oil & gas stock returns by running the following regression:

$$R_{i,j,t} = \alpha_0 + \alpha_1 MDI_{i,j,t} + \alpha_2 NYMEXreturn_{i,j,t} + \sum \alpha_3 Controls + \alpha_4 + \varepsilon_{i,j,t}$$

where $R_{i,j,t}$ is excess return of firm i , in country j , in month t over the risk-free rate; α_1 is the market dislocation index; α_2 is the NYMEX roll-on-future one month oil return; α_3 is a set of

control variables and α_4 is set of firm-fixed effects to control for unobservable time-invariant heterogeneity. We cluster standard error at the firm-level to control for intragroup correlation within the firm. The main coefficients of interests are α_1 and α_2 with the expectation that α_1 is negative and α_2 is positive. Columns 3 and 4 include control variables, whereas Columns 1 and 2 do not.

The results in Column 1 suggest that the oil & gas firms are less risky compared to overall market as the RM_RF coefficient is 0.878 and statistically significant at the one percent level. This is consistent with Boyer and Filion (2007) and Sadorsky (2001) where the authors note that oil & gas firms are less risky but poor hedging alternatives. Including MDI as an additional explanatory variable in Column 2 yields a negative coefficient of -0.0496, significant at the one percent level. This is expected as an increase in global uncertainty and stress leads to lower oil & gas stock returns. This suggests that oil & gas firms do not provide a good hedging tool when the market uncertainty and stress are high.

In Column 3, we include NYMEX oil return and other control variables. The positive effect of oil prices on oil & gas firms can be seen as the coefficient is 0.223 and statistically significant at the one percent level. This is expected as the rise of oil price increases the cash flow and profitability of oil & gas firms. The MDI coefficient remains negative and statistically significant at the one percent level. The coefficients of firm risk (standard deviation of stock returns) and illiquidity are positive, suggesting that risky and illiquidity stocks require higher return. In Column 4, we introduce additional control variables, PTBV and Leverage. Although the number of observations drops from 275,865 to 156,273, the results are consistent with

Column 3.⁵ Summing up, the result suggests that both the market uncertainty and oil price shocks are important determinants of oil & gas sector stock returns.

[Insert Table 5 here]

In Table 6, we investigate whether the effect of MDI and oil returns are felt evenly in high oil producing and net oil output countries. It can be reasoned that the countries that rely overly on oil production as a source of revenue are sensitive to oil prices and global market stability. Therefore, a change in oil prices and/or international geopolitical events can significantly impact the cash flow of the economy (Lopez-Murphy and Villafuerte, 2010). On the contrary, we argue that oil & gas stock returns in low oil producing countries are less sensitive to oil price fluctuations and global uncertainty as the economy of the country is diversified and not dependent on a single sector. To this end, we split the sample into high and low oil production/consumption countries. We use the median oil production and consumption statistics available from USEIA as a cut-off measure.

In Columns 1 and 2, we concentrate only on oil production; while in Columns 3 and 4, we concentrate on the net oil output, i.e. oil production less consumption within the country. We find that the MDI coefficient is -0.0106 (*t-stat* -2.91) for low oil producing countries compared to -0.0350 (*t-stat* -10.26) for high oil producing countries. Similarly, NYMEX oil return coefficient is 0.158 (*t-stat* 16.69) for low oil producing countries, as compared to 0.227 (*t-stat* 30.81) for high oil producing countries. Taken together, the results suggest that oil & gas firms located in high oil producing countries are more sensitive to global uncertainty and oil price shocks. Similar results can be observed when we split the sample based on net oil output. The

⁵ As the sample size drops by less than half with the inclusion of PTBV and leverage factor, we do not include them in further analysis. The results, though, remain qualitatively similar without their exclusion.

high MDI coefficient is almost three times larger than low MDI, while the NYMEX return does not significantly differ.

[Insert Table 6 here]

In this section, we examine the role of product market competition in moderating the effect of oil price shocks and market disruptions on stock returns. We form terciles based on whether the firm is located in the lowest HHI industry, the medium HHI industry or the highest HHI industry. Forming (2x1) and (4x1) vector of sales HHI dummies to capture varying degrees of competition do not alter our findings. Alternatively, results remain the same if we use competition intensity cut-offs, suggested by the US Department of Justice.⁶ The regression includes control variables and sales HHI dummies to control for the direct effect of competition on firm-level returns.

We report the results in Table 7. In Model 1, we introduce the interaction between NYMEX oil return and competition (sales HHI) tercile dummies. The NYMEXreturn coefficient in a high-competition industry, i.e., HHI (Low) is 0.310 (*t-stat* 28.36); whereas NYMEXreturn coefficient in a low-competition industry, i.e., HHI (High) is 0.166 (*t-stat* 16.55). Further, we undertake Wald test (unreported) to check if the coefficients are symmetric. The null hypothesis is rejected at the one percent level, suggesting that the oil & gas firms located in a non-competitive industry are less affected by the changes in oil price.

In Model 2, we introduce the interaction between MDI and sales HHI tercile dummies. The results are in line with Model 1, suggesting that firms located in a non-competitive industry are less sensitive to global market dislocation. In Model 3, we include the interaction between

⁶A market with a HHI of less than 0.10 is considered competitive marketplace; a result of 0.10 to 0.18 as moderately concentrated marketplace; and a result of 0.18 or greater as a highly concentrated marketplace.

NYMEX oil return, MDI and sales HHI tercile dummies. The results are consistent with Model 1 and Model 2. Wald tests (unreported) also suggest that coefficients are asymmetric. In summary, the results suggest that firms that face less competition from other firms in the oil & gas industry are less sensitive to oil price changes and market stress. These results are in line with Sen (2003), who study the retail gasoline prices in Canada. The author finds international crude input prices and product market competition (to a smaller extent) are important determinants in explaining retail gasoline prices.

[Insert Table 7 here]

Previous literature (Hamilton and Herrera, 2004; Hooker, 1996; Huang et al., 2005; Mork, 1989; Mory, 1993) suggest that the oil price shocks on economic activity are asymmetric. However, there has been little evidence on the asymmetric relationship between the oil price and stock market at the firm-level. An exception to this is Nandha and Faff (2008) who report contradictory findings and show that symmetric relationship is observed between oil prices and global industry indices. In Table 8, we present evidence of an asymmetric relationship between oil prices and oil & gas stock prices with and without controlling for competition level. We introduce two variables, an increase and decrease in oil prices, to document the impact on stock return.

In Model 1, the main coefficients of interests are Positive Oil (NYMEX oil return x positive oil price dummy) and Negative Oil (NYMEX oil return x negative oil price dummy). The results indicate that the sensitivity of firm-level returns to an increase in oil price is much lower than a decrease in oil price. We undertake a Wald test to check whether the coefficient (Positive Oil=Negative Oil) are the same. The *F*-test value is 136.82 and statistically significant at the one percent level, suggesting that coefficients are asymmetric. Similarly, we test the

hypothesis that coefficients are jointly equal to zero (Positive Oil=Negative Oil=0). The null hypothesis is again rejected (prob. <1%), suggesting the coefficients are not similar.

In Model 2, we introduce competition level in the industry and find intuitive results. We find that the increase in oil price affect the firm-level returns at the same magnitude under different levels of competition. The HHI (Low) x Positive Oil coefficient is 0.137 (*t*-stat 6.81), whereas HHI (High) x Positive Oil coefficient is 0.162 (*t*-stat 10.58). However, a significant difference is observed when we compare the oil price decrease for the monopoly firms versus the firms that face strong competition from other firms. The HHI (Low)* Negative Oil coefficient is more than double the size of HHI (High)* Negative Oil and statistically significant at the one percent level, i.e. 0.464 (*t*-stat 24.07) and 0.183 (*t*-stat 11.71) respectively.

We undertake the Wald test to check whether the coefficients are the same or jointly equal to zero. We find that the Wald test is unable to reject the null hypothesis of [HHI (Low)* Positive Oil=HHI (High)* Positive Oil] at the one percent level (*F*-value 1.01, prob. 31.57%). On the contrary, the Wald test rejects the null hypothesis of similar coefficient [HHI (Low)* Negative Oil=HHI (High)* Negative Oil] at the one percent level (*F*-value 135, prob. <1%). Taken together, the results suggest the firms that face less competition from other firms operating in the oil & gas industry do not pass on the benefit of oil price drop. Stated alternatively, the stock returns of firms operating in a low-competition industry are less affected by the decrease in oil price when compared to firms that are operating in high-competitive industries. The results are largely consistent with the “rockets and feathers” hypothesis of Bacon (1991), where the author show that the firms increase the gasoline price rapidly following an increase in the crude oil price but decreases like feathers following a drop in crude oil prices. The results are also consistent with Borenstein and Cameron (1992), who note that the asymmetry reaction of crude oil price changes to gasoline prices may arise due to the market power of some producers or distributors.

[Insert Table 8 here]

We undertake various robustness checks to reinforce the findings of this study. In Column 1 of Table 9, we follow Sadorsky (2001) and use Newey and West (1987) adjusted standard errors to account for serial correlation and heteroscedasticity. The results are consistent with our baseline regression results that are reported in Table 5. In Column 2, we use a longitudinal hierarchical model that is ideal when the data is nested in levels.⁷ This study uses firm-level data with repeated observation across years and nested within a country, the data can be organized at two levels, i.e. at the firm- and at the country-level. Time-series value of each firm can be nested at the firm-level and further at the country-level. For example, repeated monthly observations of British Petroleum are nested at the firm-level, which can be further nested within the UK (country-level). Thus, hierarchical modelling is superior in capturing interdependencies at various levels. Also, uneven sample distribution across countries is not a major concern as the hierarchical model adjusts automatically for uneven distribution of sample within the nest. We let the intercept vary at the country- and firm-level to capture interdependencies at various levels. We find the results qualitatively similar to previous results as the MDI and NYMEXret coefficients are statistically significant and in the expected direction.

[Insert Table 9 here]

The second concern that needs to be addressed is unequal sample distribution. As seen in Table 1, the sample is dominated by Canadian and US firms. Of the total 275,865 firm-month

⁷ See Searle et al. (1992); McCulloch et al. (2001); Verbeke and Molenberghs (2009); and Raudenbush and Bryk (2002) for details.

observations, Canadian and the US firms dominate with 174,775 observations, i.e., 63.36% of the full sample. Therefore, it may be the case that the Canadian and the US oil & gas firms are driving the result. To negate this concern, we run weighted regression that accounts for a number of observations associated with each country. More specifically, following Costa et al. (2013), we use the square root of a number of observations in a weighted multi-factor regression model to control for uneven sample size across countries. The benefit of using square root of a number of observations is that standard error of the mean is a function of the square root of n . We find the magnitude and sign of the interested coefficients similar to our baseline regressions. We also consider excluding the firms listed in Canada and the US altogether from the regression, but our major findings do not weaken, as reported in Column 4 of the table.

5. Conclusion

In this study, we propose that systematic asset price risk of oil & gas firms are better modelled if both oil price shocks and macroeconomic forces are taken into consideration. Previous studies use various measures of macroeconomic forces, such as USD/CAD exchange rate, local interest rates. However, as oil prices are not country-specific, a better proxy is needed that measures market stress and uncertainty. Further, the use of sub-index as a performance benchmark of oil & gas firms may skew the returns positively and make the statistical inferences unreliable. To this end, we use a better measure of market stress and undertake a comprehensive analysis using firm-level data from 72 countries spanning three decades.

We find that both oil price shocks and market stress have a strong impact on firm-level stock returns. The firm-level stock returns are positively and negatively related to the oil price and market stress respectively. We also investigate the role of country-level determinants in explaining the sensitivity of firm-level stock returns towards oil price and market stress. We

find that firms located in oil-rich countries are more sensitive to oil price fluctuation and market stress than firms located in less oil producing countries.

We next investigate the role of product market competition on oil & gas stock returns. To determine the competition level in the industry, we compute sales HHI index for oil & gas sector for each country and for each year. We find that firms that face less competition are less sensitive to oil price changes and market stress than those that face strong competition from other firms. The results suggest that the market power of the firms dampen the volatility of the firm-level returns. Finally, we consider the asymmetric effect of oil price on firm-level returns. We find that stock returns of firms operating in a low-competition industry are less affected by the decrease in oil price when compared to firms that face strong competition from other firms.

References

- Bacon, R.W., 1991. Rockets and feathers: the asymmetric speed of adjustment of UK retail gasoline prices to cost changes. *Energy Economics* 13, 211-218.
- Banz, R.W., Breen, W.J., 1986. Sample-dependent results using accounting and market data: Some evidence. *Journal of Finance* 41, 779-793.
- Borenstein, S., Cameron, A.C., 1992. Do gasoline prices respond asymmetrically to crude oil price changes? *Quarterly Journal of Economics*, pp. 305-339.
- Boyer, M.M., Filion, D., 2007. Common and fundamental factors in stock returns of Canadian oil and gas companies. *Energy Economics* 29, 428-453.
- Brown, S.J., Goetzmann, W., Ibbotson, R.G., Ross, S.A., 1992. Survivorship bias in performance studies. *Review of Financial Studies* 5, 553-580.
- Burbidge, J., Harrison, A., 1984. Testing for the effects of oil-price rises using vector autoregressions. *International Economic Review*, 459-484.
- Chen, N.-F., Roll, R., Ross, S.A., 1986. Economic forces and the stock market. *Journal of Business*, 383-403.
- Chui, A.C., Titman, S., Wei, K.J., 2010. Individualism and momentum around the world. *Journal of Finance* 65, 361-392.
- Costa, B., Crawford, A., Jakob, K., 2013. Does culture influence IPO underpricing? *Journal of Multinational Financial Management* 23, 113-123.
- Deltas, G., 2008. Retail gasoline price dynamics and local market power. *Journal of Industrial Economics* 56, 613-628.
- El-Sharif, I., Brown, D., Burton, B., Nixon, B., Russell, A., 2005. Evidence on the nature and extent of the relationship between oil prices and equity values in the UK. *Energy Economics* 27, 819-830.
- Elliott, G., Rothenberg, T.J., Stock, J.H., 1996. Efficient tests for an autoregressive unit root. *Econometrica: Journal of the Econometric Society*, 813-836.
- Elyasiani, E., Mansur, I., Odusami, B., 2011. Oil price shocks and industry stock returns. *Energy Economics* 33, 966-974.
- Fama, E., French, K., 1992. The cross-section of expected stock returns. *Journal of Finance* 47, 427-465.
- Fama, E., French, K., 1997. Industry costs of equity. *Journal of Financial Economics* 43, 153-193.

- Gisser, M., Goodwin, T.H., 1986. Crude oil and the macroeconomy: Tests of some popular notions: Note. *Journal of Money, Credit and Banking*, 95-103.
- Hamilton, J.D., 1983. Oil and the macroeconomy since World War II. *Journal of Political Economy*, 228-248.
- Hamilton, J.D., 2003. What is an oil shock? *Journal of Econometrics* 113, 363-398.
- Hamilton, J.D., Herrera, A., 2004. Oil shocks and aggregate macroeconomic behavior: The role of monetary policy. *Journal of Money, Credit, and Banking* 36, 265.
- Hooker, M.A., 1996. What happened to the oil price-macroeconomy relationship? *Journal of Monetary Economics* 38, 195-213.
- Huang, B.-N., Hwang, M., Peng, H.-P., 2005. The asymmetry of the impact of oil price shocks on economic activities: An application of the multivariate threshold model. *Energy Economics* 27, 455-476.
- Huang, R.D., Masulis, R.W., Stoll, H.R., 1996. Energy shocks and financial markets. *Journal of Futures Markets* 16, 1-27.
- Ince, O., Porter, R., 2006. Individual equity return data from Thomson Datastream: Handle with care! *Journal of Financial Research* 29, 463-479.
- Jones, C.M., Kaul, G., 1996. Oil and the stock markets. *Journal of Finance* 51, 463-491.
- Kilian, L., Park, C., 2009. The impact of oil price shocks on the us stock market. *International Economic Review* 50, 1267-1287.
- Kwiatkowski, D., Phillips, P.C., Schmidt, P., Shin, Y., 1992. Testing the null hypothesis of stationarity against the alternative of a unit root: How sure are we that economic time series have a unit root? *Journal of Econometrics* 54, 159-178.
- Lesmond, D., Ogden, J., Trzcinka, C., 1999. A new estimate of transaction costs. *Review of Financial Studies* 12, 1113-1141.
- Lopez-Murphy, P., Villafuerte, M., 2010. Fiscal policy in oil producing countries during the recent oil price cycle. *IMF Working Papers*, 1-23.
- Miller, J.I., Ratti, R.A., 2009. Crude oil and stock markets: Stability, instability, and bubbles. *Energy Economics* 31, 559-568.
- Mork, K.A., 1989. Oil and the macroeconomy when prices go up and down: an extension of Hamilton's results. *Journal of Political Economy*, 740-744.
- Mory, J.F., 1993. Oil prices and economic activity: Is the relationship symmetric? *Energy Journal*, 151-161.
- Nandha, M., Faff, R., 2008. Does oil move equity prices? A global view. *Energy Economics* 30, 986-997.

- Newey, W., West, K., 1987. Hypothesis testing with efficient method of moments estimation. *International Economic Review* 28, 777-787.
- Oladunjoye, O., 2008. Market structure and price adjustment in the US wholesale gasoline markets. *Energy Economics* 30, 937-961.
- Papapetrou, E., 2001. Oil price shocks, stock market, economic activity and employment in Greece. *Energy Economics* 23, 511-532.
- Park, J., Ratti, R.A., 2008. Oil price shocks and stock markets in the US and 13 European countries. *Energy Economics* 30, 2587-2608.
- Pasquariello, P., 2014. Financial market dislocations. *Review of Financial Studies* 27, 1868-1914.
- Phillips, P.C., Perron, P., 1988. Testing for a unit root in time series regression. *Biometrika* 75, 335-346.
- Raudenbush, S.W., Bryk, A.S., 2002. *Hierarchical linear models: Applications and data analysis methods*. Sage.
- Sadorsky, P., 1999. Oil price shocks and stock market activity. *Energy Economics* 21, 449-469.
- Sadorsky, P., 2001. Risk factors in stock returns of Canadian oil and gas companies. *Energy Economics* 23, 17-28.
- Sen, A., 2003. Higher prices at Canadian gas pumps: international crude oil prices or local market concentration? An empirical investigation. *Energy Economics* 25, 269-288.
- Tirole, J., 1988. *The theory of industrial organization*. MIT press.
- Wen, X., Wei, Y., Huang, D., 2012. Measuring contagion between energy market and stock market during financial crisis: A copula approach. *Energy Economics* 34, 1435-1446.

Table 1: Sample description

No	Country	Firm-Months	Country-Months	Production	Consumption
1	Argentina	962	322	801.66	541.57
2	Australia	23,530	380	589.64	988.95
3	Austria	323	323	26.71	261.56
4	Bangladesh	818	271	5.68	85.70
5	Belgium	160	160	10.91	550.00
6	Brazil	638	244	2,283.50	2,296.55
7	Bulgaria	135	135	3.38	99.45
8	Canada	120,384	380	2,953.70	2,065.07
9	Chile	305	305	17.94	248.50
10	China	1,321	252	3,809.02	6,795.44
11	Colombia	299	242	690.28	276.97
12	Cote d'Ivoire	156	78	NA	NA
13	Cyprus	306	167	0.02	56.00
14	Czech Republic	426	236	10.28	178.75
15	Ecuador	196	196	397.66	139.97
16	Egypt	189	111	711.69	712.19
17	Finland	147	147	9.79	207.67
18	France	2,280	380	97.28	1,915.00
19	Germany	578	314	139.87	2,533.45
20	Ghana	64	64	77.79	64.73
21	Greece	387	198	6.63	408.39
22	Hong Kong	248	166	0.10	354.00
23	Hungary	294	228	37.00	145.00
24	Iceland	63	63	NA	18.31
25	India	6,986	299	818.33	2,426.33
26	Indonesia	722	241	1,041.98	1,341.34
27	Ireland	2,580	368	-0.23	143.22
28	Israel	4,199	263	3.91	242.15
29	Italy	630	228	154.51	1,728.68
30	Japan	5,974	379	104.45	5,318.76
31	Jordan	178	102	0.16	107.73
32	Kazakhstan	157	68	1,608.69	216.00
33	Kuwait	190	111	2,691.82	397.44
34	Lithuania	112	112	9.83	56.53
35	Luxembourg	104	46	NA	59.01
36	Malaysia	1,264	347	740.75	475.10
37	Mauritius	76	76	NA	22.00
38	Morocco	411	225	4.31	174.14
39	Netherlands	916	379	74.88	769.00
40	New Zealand	397	323	52.11	131.00
41	Nigeria	397	64	2,459.40	287.42
42	Norway	1,885	379	2,352.56	221.00
43	Oman	109	109	818.89	113.55
44	Pakistan	2,555	269	66.59	360.12
45	Papua New Guinea	479	380	55.15	15.45
46	Peru	191	191	120.50	165.72
47	Philippines	2,081	326	14.10	311.67
48	Poland	578	180	33.84	527.02
49	Portugal	98	98	5.42	273.69
50	Qatar	10	10	NA	NA
51	Romania	631	197	116.64	219.61
52	Russia	5,133	203	9,732.35	2,803.47
53	Saudi Arabia	193	110	11,428.60	2,579.73
54	Singapore	1,297	380	8.57	743.87
55	Slovak Republic	90	90	10.30	78.51
56	Slovenia	191	191	0.10	53.64
57	South Africa	722	380	201.41	465.67
58	South Korea	2,532	327	16.92	2,155.12
59	Spain	599	333	29.49	1,385.32
60	Sri Lanka	217	217	-0.54	81.29
61	Sweden	2,444	306	7.27	350.38
62	Switzerland	316	208	3.61	265.63
63	Syria	45	45	170.53	312.00
64	Taiwan	132	132	21.21	960.45
65	Thailand	1,822	334	309.53	1,007.58
66	Turkey	1,158	323	57.63	647.50
67	Ukraine	278	104	93.56	287.43
68	United Arab Emirates	57	57	2,947.50	585.64
69	United Kingdom	15,167	380	90.46	1,746.99
70	United States	54,391	380	9,444.55	18,771.40
71	Vietnam	777	96	339.76	431.76
72	Yugoslavia	185	68	21.22	79.85
Total		275,865	15,796		

Table 2: Summary statistics of main variables

Variable	Mean	StDev	Median	Skewness	Kurtosis	N
Stock return	-0.56%	20.89%	0.00%	0.09	7.95	275,865
Market return	0.60%	5.20%	1.05%	-0.49	13.36	275,865
MDI	-0.09	0.19	-0.10	2.49	15.75	275,865
NYMEXret	0.34%	9.31%	0.92%	-0.34	4.95	275,865
St Dev of Return	5.46%	4.93%	3.92%	2.23	9.80	275,865
Illiquidity	39.06%	30.00%	30.53%	0.63	2.14	275,865
Log of MV	3.66	3.06	3.50	0.23	2.56	275,865
PTBV	2.75	4.42	1.69	6.28	53.76	167,261
Leverage	12.77%	16.05%	4.97%	1.23	3.83	166,480
Riskfree	0.46%	0.22%	0.44%	0.71	4.48	275,865

In this table we report distribution of main variables. Stock return is monthly log firm-level return, calculated as $R_{i,t} = \ln(P_t/P_{t-1})$, where P is the price of stock i in month t . Market return is monthly log return of the market in which the firm is located. MDI is based on the law of one price and derived from the deviation in the Covered Interest Rate Parity (CIRP), Triangular Arbitrage Parity (TAP) and American Depositary Receipt Parity (ADRP). NYMEX is New York Mercantile Exchange roll-on-future one month oil price. Riskiness of firm is computed as daily standard deviation of stock returns over the last one year. Illiquidity is calculated as per the Lesmond et al. (1999) model, where a stock with no change in price over a period of time is considered illiquid. Log of MV is logarithm of market value in US\$. PTBV is price to book value and leverage is long-term debt divided by total assets. Risk-free rate are country-specific monthly rates retrieved from Datastream.

Table 3: Correlation coefficient

	Stock return	Market return	MDI	NYMEXret	St Dev of Return	Illiquidity	Log of MV	PTBV	Leverage	Riskfree
Stock return	1									
Market return	0.213***	1								
MDI	-0.0882***	-0.236***	1							
NYMEXret	0.141***	0.209***	-0.229***	1						
St Dev of Return	-0.00194	-0.00134	0.0918***	0.00431*	1					
Illiquidity	-0.00176	0.00585**	0.0245***	0.0163***	0.288***	1				
Log of MV	0.0731***	0.00522**	-0.0760***	-0.00152	-0.548***	-0.663***	1			
PTBV	0.0546***	0.0124***	-0.0256***	0.0134***	0.0403***	0.0369***	-0.0292***	1		
Leverage	0.00750**	0.0114***	-0.00585*	-0.0192***	-0.207***	-0.298***	0.336***	-0.0180***	1	
Riskfree	0.00546**	-0.00849***	0.0636***	0.00700***	0.0301***	0.0460***	-0.118***	-0.0330***	0.116***	1

In this table we report correlation coefficient between main variables. Stock return is monthly log firm-level return, calculated as $R_{i,t} = \ln(P_t/P_{t-1})$, where P is the price of stock i in month t . Market return is monthly log return of the market in which the firm is located. MDI is based on the law of one price and derived from the deviation in the Covered Interest Rate Parity (CIRP), Triangular Arbitrage Parity (TAP) and American Depositary Receipt Parity (ADRP). NYMEX is New York Mercantile Exchange roll-on-future one month oil price. Riskiness of firm is computed as daily standard deviation of stock returns over the last one year. Illiquidity is calculated as per the Lesmond et al. (1999) model, where a stock with no change in price over a period of time is considered illiquid. Log of MV is logarithm of market value in US\$. PTBV is price to book value and leverage is long-term debt divided by total assets. Risk-free rate are country-specific monthly rates retrieved from Datastream. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level.

Table 4: Unit root test

MDI			NYMEX		Log dif NYMEX	
Lags	DF-GLS tau statistic	1% critical value	DF-GLS tau statistic	1% critical value	DF-GLS tau statistic	1% critical value
4	-5.687	-3.480	-2.562	-3.480	-7.984	-3.480
3	-6.168	-3.480	-2.635	-3.480	-8.344	-3.480
2	-6.232	-3.480	-2.569	-3.480	-9.100	-3.480
1	-6.353	-3.480	-2.451	-3.480	-11.741	-3.480

In this table we present a modified Dickey-Fuller t -test, proposed by Elliott et al. (1996) to check if the unit root problem exists.

Table 5: Effect of market dislocation and oil price shocks on oil & gas prices

	(1) RI_RF	(2) RI_RF	(3) RI_RF	(4) RI_RF
RM_RF	0.878*** (58.09)	0.834*** (54.46)	0.764*** (49.78)	0.851*** (61.39)
MDI		-0.0496*** (-20.47)	-0.0161*** (-6.54)	-0.0170*** (-6.24)
NYMEXret			0.223*** (36.88)	0.259*** (34.50)
St. Dev of Ret			0.514*** (30.08)	0.655*** (18.33)
Illiquidity			0.134*** (28.25)	0.138*** (18.58)
Log of MV			0.0274*** (30.67)	0.0237*** (20.41)
PTBV				0.00325*** (11.05)
Leverage				-0.0421*** (-6.34)
Constant	-0.0114*** (-572.08)	-0.0158*** (-75.13)	-0.194*** (-37.39)	-0.198*** (-24.49)
Firm F.E.?	Yes	Yes	Yes	Yes
Clustered S.E.?	Firm-level	Firm-level	Firm-level	Firm-level
N	293990	293990	275865	156273
Adj. R-sq	0.045	0.047	0.080	0.136

In this table we investigate the impact of oil returns and market dislocation on the oil & gas stock returns. The dependent variable is stock return less risk-free return (RI_RF), calculated as $R_{i,t} - R_{f,t}$. Risk-free adjusted market return (RM_RF) is monthly log return of the market in which the firm is located less corresponding risk-free return. MDI is based on the law of one price and derived from the deviation in the Covered Interest Rate Parity (CIRP), Triangular Arbitrage Parity (TAP) and American Depositary Receipt Parity (ADRP). NYMEXret is New York Mercantile Exchange roll-on-future one month oil price change. Riskiness of firm is computed as daily standard deviation of stock returns over the last one year. Illiquidity is calculated as per the Lesmond et al. (1999) model, where a stock with no change in price over a period of time is considered illiquid. Log of MV is logarithm of market value in US\$. PTBV is price to book value and leverage is long-term debt divided by total assets. The model includes firm-fixed effects. Standard error are clustered at the firm-level. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. *t-stats* are given in parenthesis and are based on robust standard errors.

Table 6: Country-level production and consumption of oil

	(1)	(2)	(3)	(4)
	Production		Production-Consumption	
	Low	High	Low	High
RM_RF	0.815*** (37.63)	0.718*** (34.95)	0.807*** (44.37)	0.709*** (29.35)
MDI	-0.0106*** (-2.91)	-0.0350*** (-10.26)	-0.0145*** (-4.51)	-0.0356*** (-9.05)
NYMEXret	0.158*** (16.69)	0.227*** (30.81)	0.221*** (25.85)	0.189*** (23.48)
St. Dev of Ret	0.545*** (11.30)	0.539*** (27.25)	0.498*** (12.43)	0.557*** (27.06)
Illiquidity	0.107*** (14.69)	0.149*** (23.94)	0.117*** (17.59)	0.155*** (22.60)
Log of MV	0.0243*** (19.18)	0.0300*** (26.53)	0.0252*** (22.91)	0.0307*** (22.55)
Constant	-0.178*** (-20.89)	-0.206*** (-32.70)	-0.184*** (-24.30)	-0.208*** (-30.26)
Firm F.E.?	Yes	Yes	Yes	Yes
Clustered S.E.?	Firm-level	Firm-level	Firm-level	Firm-level
N	90577	173234	126128	137683
Adj. R-sq	0.113	0.071	0.113	0.066

In this table we investigate whether the effect of MDI and oil returns are felt evenly in high oil producing and net oil output countries. We use the median oil production and consumption statistics available from USEIA as a cut-off measure. The dependent variable is stock return less risk-free return (RI_RF), calculated as $R_{i,t} - R_{f,t}$. Risk-free adjusted market return (RM_RF) is monthly log return of the market in which the firm is located less corresponding risk-free return. MDI is based on the law of one price and derived from the deviation in the Covered Interest Rate Parity (CIRP), Triangular Arbitrage Parity (TAP) and American Depositary Receipt Parity (ADRP). NYMEXret is New York Mercantile Exchange roll-on-future one month oil price change. Riskiness of firm is computed as daily standard deviation of stock returns over the last one year. Illiquidity is calculated as per the Lesmond et al. (1999) model, where a stock with no change in price over a period of time is considered illiquid. Log of MV is logarithm of market value in US\$. PTBV is price to book value and leverage is long-term debt divided by total assets. The model includes firm-fixed effects. Standard error are clustered at the firm-level. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. t-stats are given in parenthesis and are based on robust standard errors.

Table 7: Effect of competition on oil & gas stock prices

	(1)	(2)	(3)
	RI_RF	RI_RF	RI_RF
RM_RF	0.740*** (47.41)	0.753*** (48.40)	0.741*** (47.37)
HHI(Low) x NYMEXret	0.310*** (28.36)		0.300*** (27.09)
HHI(Medium) x NYMEXret	0.178*** (18.14)		0.178*** (18.10)
HHI(High) x NYMEXret	0.166*** (16.55)		0.172*** (17.27)
HHI(Low) x MDI		-0.0456*** (-11.15)	-0.0319*** (-7.84)
HHI(Medium) x MDI		-0.0151*** (-2.62)	-0.0167*** (-2.87)
HHI(High) x MDI		-0.00635* (-1.66)	-0.0127*** (-3.43)
MDI	-0.0220*** (-8.52)		
NYMEXret		0.209*** (33.80)	
St. Dev of Ret	0.527*** (28.79)	0.528*** (28.72)	0.527*** (28.76)
Illiquidity	0.149*** (26.90)	0.149*** (26.88)	0.149*** (26.86)
Log of MV	0.0310*** (30.39)	0.0309*** (30.32)	0.0310*** (30.36)
HHI(Medium) Dummy	0.0258*** (13.26)	0.0270*** (13.67)	0.0267*** (13.49)
HHI(High) Dummy	0.0222*** (6.75)	0.0230*** (6.99)	0.0234*** (7.09)
Constant	-0.228*** (-35.13)	-0.229*** (-35.22)	-0.229*** (-35.21)
Firm F.E.?	Yes	Yes	Yes
Clustered Standard Error?	Firm-level	Firm-level	Firm-level
N	248175	248175	248175
Adj. R-sq	0.081	0.081	0.081

In this table we examine the role of product market competition in moderating the effect of oil price shocks and market disruptions on stock returns. The dependent variable is stock return less risk-free return (RI_RF), calculated as $R_{i,t} - R_{f,t}$. Risk-free adjusted market return (RM_RF) is monthly log return of the market in which the firm is located less corresponding risk-free return. MDI is based on the law of one price and derived from the deviation in the Covered Interest Rate Parity (CIRP), Triangular Arbitrage Parity (TAP) and American Depositary Receipt Parity (ADRP). NYMEXret is New York Mercantile Exchange roll-on-future one month oil price change. We form terciles on the basis of whether the firm is located in the lowest HHI industry, the medium HHI industry or the highest HHI industry. Riskiness of firm is computed as daily standard deviation of stock returns over the last one year. Illiquidity is calculated as per the Lesmond et al. (1999) model, where a stock with no change in price over a period of time is considered illiquid. Log of MV is logarithm of market value in US\$. PTBV is price to book value and leverage is long-term debt divided by total assets. The model includes firm-fixed effects. Standard error are clustered at the firm-level. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. t-stats are given in parenthesis and are based on robust standard errors.

Table 8: Asymmetric effect of oil price

	(1) RI_RF	(2) RI_RF
RM_RF	0.768*** (50.40)	0.742*** (47.47)
MDI	-0.00487* (-1.93)	-0.0104*** (-3.91)
Positive Oil	0.132*** (13.19)	
Negative Oil	0.312*** (32.48)	
HHI(Low) x Positive Oil		0.137*** (6.81)
HHI(Medium) x Positive Oil		0.129*** (8.00)
HHI(High) x Positive Oil		0.162*** (10.58)
HHI(Low) x Negative Oil		0.464*** (24.07)
HHI(Medium) x Negative Oil		0.230*** (16.02)
HHI(High) x Negative Oil		0.183*** (11.71)
St. Dev of Return	0.516*** (30.12)	0.527*** (28.73)
Illiquidity	0.134*** (28.21)	0.150*** (26.88)
Log of MV	0.0274*** (30.76)	0.0311*** (30.45)
HHI(Medium) Dummy		0.0182*** (7.78)
HHI(High) Dummy		0.0122*** (3.34)
Constant	-0.187*** (-36.09)	-0.217*** (-33.34)
Firm F.E.?	Yes	Yes
Clustered Standard Error?	Firm-level	Firm-level
N	275865	248175
Adj. R-sq	0.081	0.082

In this table we present evidence of an asymmetric relationship between oil prices and oil & gas stock prices with and without controlling for competition level. The dependent variable is stock return less risk-free return (RI_RF), calculated as $R_{i,t} - R_{f,t}$. Risk-free adjusted market return (RM_RF) is monthly log return of the market in which the firm is located less corresponding risk-free return. MDI is based on the law of one price and derived from the deviation in the Covered Interest Rate Parity (CIRP), Triangular Arbitrage Parity (TAP) and American Depositary Receipt Parity (ADRP). NYMEXret is New York Mercantile Exchange roll-on-future one month oil price change. We form terciles on the basis of whether the firm is located in the lowest HHI industry, the medium HHI industry or the highest HHI industry. We introduce two variables, an increase and decrease in oil prices, to document the impact on stock return. Riskiness of firm is computed as daily standard deviation of stock returns over the last one year. Illiquidity is calculated as per the Lesmond et al. (1999) model, where a stock with no change in price over a period of time is considered illiquid. Log of MV is logarithm of market value in US\$. PTBV is price to book value and leverage is long-term debt divided by total assets. The model includes firm-fixed effects. Standard error are clustered at the firm-level. * statistically significant at the 10 percent, ** statistically significant at the 5 percent and *** statistically significant at the 1 percent level. t-stats are given in parenthesis and are based on robust standard errors.

Table 9: Robustness checks

	Newey-West S.E Adjusted (1) RI_RF	Hierarchical Model (2) RI_RF	Weighted Regression (3) RI_RF	Excluding Canada and the US (4) RI_RF
RM_RF	0.763*** (88.37)	0.764*** (99.51)	0.822*** (51.23)	0.737*** (30.63)
MDI	-0.0199*** (-8.12)	-0.0180*** (-8.52)	-0.0228*** (-6.51)	-0.0101*** (-2.92)
NYMEXret	0.229*** (49.31)	0.226*** (52.91)	0.233*** (30.39)	0.163*** (17.51)
St. Dev of Ret	0.289*** (20.11)	0.418*** (40.04)	0.529*** (27.19)	0.543*** (12.99)
Illiquidity	0.0570*** (31.34)	0.100*** (46.12)	0.157*** (24.31)	0.102*** (15.12)
Log of MV	0.0113*** (55.74)	0.0192*** (67.13)	0.0300*** (23.39)	0.0236*** (20.12)
Constant	-0.0935*** (-50.03)	-0.161*** (-37.55)	-0.202*** (-31.27)	-0.178*** (-22.32)
N	275865	275865	275865	101090
Adj. r-sq			0.0691	0.105

In this table we undertake a battery of robustness checks. In Column 1, we follow Sadorsky (2001) and use Newey and West (1987) adjusted standard errors to account for serial correlation and heteroscedasticity. In Column 2, we use a longitudinal hierarchical model that is ideal when the data is nested in levels. In Column 3, we run weighted regression that accounts for a number of observations associated with each country. In Column 4, we exclude the firms listed in Canada and the US altogether from the regression.

Figure 1: Market Dislocation Index (MDI) and NYMEX Oil Price Index

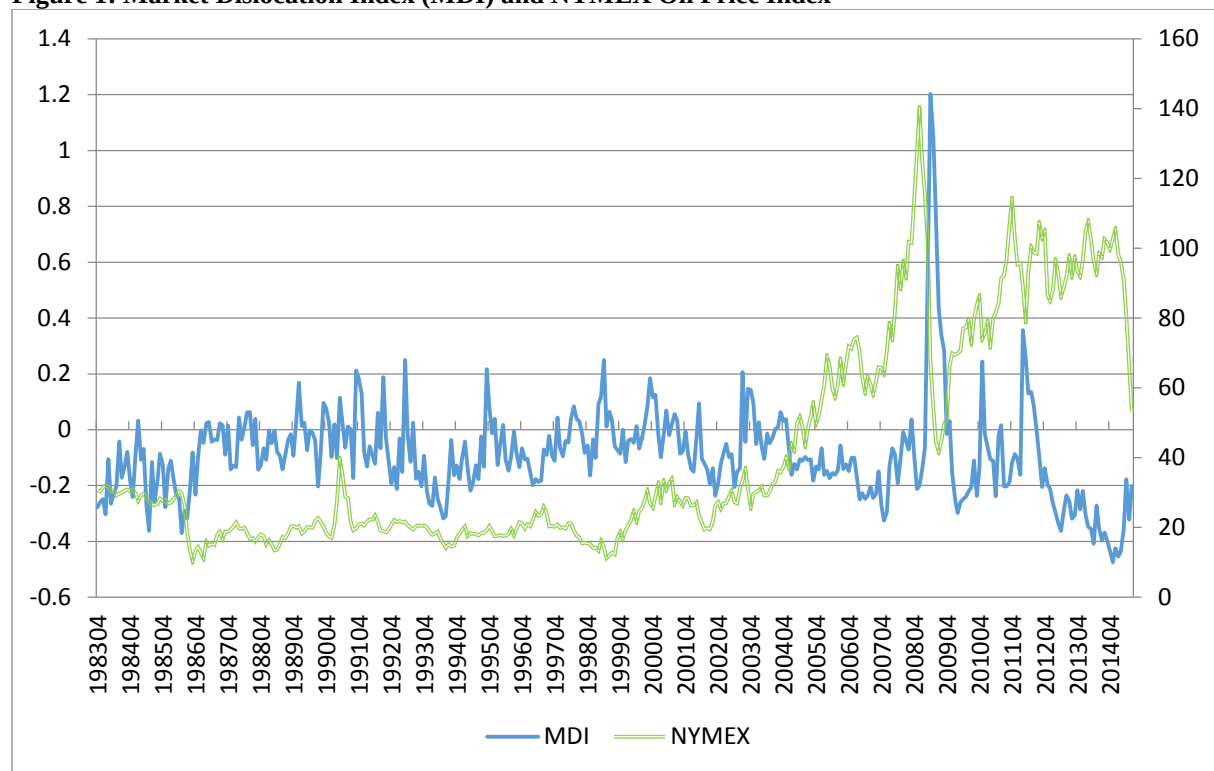


Figure 2: NYMEX Oil Price, Global Oil & Gas Industry Index and Global Market Index

